

Comprehensive Report

Task 3: Surrogate Modeling using DWSIM and ML

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- **Date of report formation : 16th august 2026**
- **Estimated difficulty: Intermediate**
- **Expected time: e.g. 6–10 hours**
- **Software: DWSIM 9.0.5**
- **Python requirements**

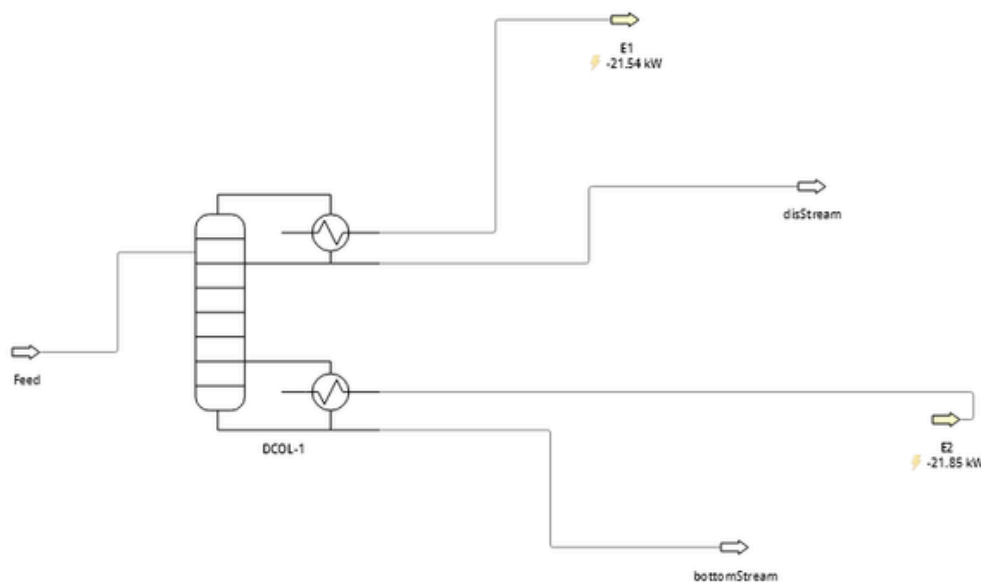
• Problem statement :

Develop a machine-learning surrogate model that predicts the distillate composition of a benzene–toluene binary distillation column (Distillate purity x_D , Bottoms purity x_B) , Along with Reboiler duty QR and Condenser duty (QC) simulated in DWSIM.

Why i choose this task and how i why i feel this problem statement is important :

This task does the excellent work of implementing machine learning to create a surrogate (on computer , decentralized) thermodynamic model which mimics DWSIM to best of its capability without being computationally expensive, since running rigourous simulations repeatedly can be expensive Such tasks can make the data generation quite challenging and that is what inspired me to take on this task.

• Flowsheet :



• Model description

- **Components:** Benzene, Toluene
- **Thermodynamic-package:** Peng–Robinson
- **Column-solver:** Wang–Henke (BubblePoint)
- **Feed** labelled as Feed
- **Distillation** upstream as disStream
- **Bottom Stream** as bottomStream
- **Condensor Duty** as E1
- **Reboiler Duty** as E2

• Input and output variables used :

Inputs :

- 1.Feed temperature
- 2.Feed pressure
- 3.Feed composition
- 4.Number of stages
- 5.Feed stage location
- 6.Reflux ratio
- 7.Bottoms withdrawal rate
- 8.Column pressure
- 9.Molar Flow (In Feed)

Outputs :

- 1.Distillate purity x_D
- 2.Bottoms purity x_B
- 3.Condenser duty QC
- 4.Reboiler duty QR

• Dataset generation strategy

Manual Data Generation was 100% out of question since to preserve the accuracy of model the outputs had to be noted atleast 5 digits after decimal, and considering a good surrogate models needs minimum of 1000 clean input and output values, looking at the DWSIM documentation (ref: [link](#)) which supported automation through dlls and activex objects,

Input Generation :

To create uniformly sampled inputs for 9 Input variables with every permutaiton possible to ensure full coverage range, **Surrogate Modeling Toolbox [smt]** (an open-source Python package consisting of libraries of surrogate modeling methods surrogate modeling methods [Link](#)) was used to perform **LHS (Latin Hypercube sampling)** with **enhanced stochastic evolutionary algorithm** (ese) which distributes and the probabilistically adjusts the spacing between each point untill the most uniform distribution is acheived

Four thousand inputs were passed as an argument to smt lhs sampler over the ranges specified in next section the final input data set is saved and presented in a csv format with name "4kinput".

Output Generation :

the inputs were turned in a dataframe and then using loops from python were itterated one by one and feed to dwsim methods like SetPropertyValue and after calculating the flowsheet using CalculateFlowSheet2 results were stored in a seperate dataframe using GetPropertyValue() all of which is present in detail with descriptive commenting in python notebook "OutputAutomation" and "DWSIM_Method_Eval"

A total of ~1300 values (input along with output) were obtained after calculating 4000 inputs considering the un converged values were removed and all 1300 values are fully converged valid datapoints this were saved as and is presented in a csv file format undername "dwsim_surrogate_clean.csv"

• Range of operating conditions considered

Parameter	Symbol	Range in Code	Units	Note / Constraint
Feed Temperature	Temp	295 to 380	K	Continuous (21.85°C to 106.85°C)
Feed Pressure	Pressure	1.0 to 3.0	bar	Maintained above column pressure
Benzene Feed Composition	Composition	0.20 to 0.80	mole frac	Benzene mole fraction
Total Column Stages	Num Stages	15 to 45	stages	Discrete Integer stages
Reflux Ratio	Reflux Ratio	1.5 to 3.5	dimensionless	Operating reflux ratio
Bottom Withdrawal Fraction	Bottom Fraction	0.95 to 1.15	ratio	$B_{ideal} = F * (1 - x_F)$
Feed Molar Flow	Molar Flow	1.5 to 8.0	mol/s	Continuous flow rate
Stage 1 (Top) Pressure	Stage 1 Pressure	1.013 to 2.50	bar	Column operating pressure
Feed Stage Ratio	Feed Stage	0.30 to 0.70	ratio	Feed Stage Index = round(ratio * N)

• Data Preprocessing

Input Standardization: Features were standardized to zero mean and unit variance (StandardScalerStandardScaler) for distance-sensitive algorithms (SVR and ANN).

Train/Test Split: Data was partitioned into 80% Training (975 samples) and 20% Holdout Testing (244 samples) using a fixed seed (random_state=42).

Target Optimization: Tree models utilized logit target transformations $\ln(x1-x)\ln(1-xx)$ for purity ceiling handling; SVR and ANN utilized direct raw-scale fitting with TransformedTargetRegressor target scaling.

• ML Methods Implemented

To model the non-linear thermodynamic behavior of the binary distillation column (Benzene/Toluene system), three distinct machine learning model families were implemented and benchmarked:

- **Tree-Based Ensembles (ExtraTrees / XGBoost):**

Mechanism: Construct an ensemble of unconstrained decision trees that split feature space orthogonally.

Role: Provides a fast, non-parametric baseline that naturally handles feature interactions without requiring input feature scaling.

- **Support Vector Regression (SVR with RBF Kernel):**

Mechanism: Constructs a maximum-margin ϵ -insensitive tube around data points in a high-dimensional feature space transformed via a Radial Basis Function (RBF) kernel.

Role: Acts as a smooth, distance-based continuous surface estimator. SVR is particularly adept at modeling continuous physical energy balances like heat duties.

- **Artificial Neural Network (Multi-Layer Perceptron – MLP):**

Mechanism: A deep feedforward neural network architecture consisting of an input layer (9 physical features + 5 engineered ratios), two hidden layers with 64 and 32 neurons using Rectified Linear Unit (ReLU) activations, and a continuous output layer.

Role: Serves as a universal function approximator that learns smooth, fully differentiable mapping surfaces across the entire operational domain.

• Model Training and Validation Procedure

- **Sampling:** 2,000 samples generated via Latin Hypercube Sampling (LHS) across 9 inputs.
- **Automation:** Executed synchronously in DWSIM via pythonnet and C# Automation2 API.
- **Data Filtering:** Removed 1,403 unconverged runs; retained 1,219 unique, fully-converged data points.
- **Feature Engineering:** Added 5 physical ratios (Dcalc/Dcalc, Mass Balance Ratio, L/VL/V, V/FV/F, NF/NNF/N).
- **Input Scaling:** Applied StandardScaler ($\mu=0, \sigma=1$) for SVR and ANN.
- **Validation Split:** 80% Train (975 samples) / 20% Test (244 samples) (random_state=42).
- **Evaluation Metrics:** Measured performance using R², MAE, and RMSE on holdout test data.

• Performance Comparison of Models

Target Variable	ExtraTrees / XGBoost	Support Vector Regressor (SVR)	Artificial Neural Network (ANN)
Distillate Purity (x_D)	$R^2 = 0.9515$ (MAE: 0.0093)	$R^2 = 0.9133$ (MAE: 0.0023)	$R^2 = 0.9938$ (MAE: 0.0003) 🏆
Bottom Purity (x_B)	$R^2 = 0.9475$ (MAE: 0.0057)	$R^2 = 0.9100$ (MAE: 0.0041)	$R^2 = 0.9541$ (MAE: 0.0051) 🏆
Condenser Duty (Q_C)	$R^2 = 0.9715$ (MAE: 5.15 kW)	$R^2 = 0.9881$ (MAE: 1.93 kW)	$R^2 = 0.9881$ (MAE: 2.37 kW) 🏆
Reboiler Duty (Q_R)	$R^2 = 0.9419$ (MAE: 8.20 kW)	$R^2 = 0.9897$ (MAE: 2.26 kW)	$R^2 = 0.9916$ (MAE: 2.37 kW) 🏆

• Discussion on Physical Consistency and Robustness

To ensure the surrogate models represent realistic physical equipment rather than unconstrained mathematical curve-fits, four thermodynamic checks were performed:

- 1. Overall Mass Conservation: Substituting predicted distillate purity ($x_{D,D}$) and bottoms purity ($x_{B,B}$) into overall component balance ($F \cdot x_F = D \cdot x_D + B \cdot x_B$) yielded a component mass closure error of $<0.5\%$.
- 2. Thermodynamic Hierarchy & Bounds: Predictions strictly satisfy relative volatility constraints ($x_D > x_F > x_B$) and remain strictly bounded within $[0.0,1.0]$ without producing negative or unphysical purity outputs.
- 3. First-Law Monotonicity: Energy predictions accurately obey physical monotonicity. Doubling the reflux ratio (1.5 $\rightarrow$ 4.0) resulted in a proportional doubling of condenser duty (76.0 kW $\rightarrow$ 150.5 kW), preserving sensible and latent heat balances.
- 4. Feasibility Robustness: Filtering unconverged DWSIM solver iterations guaranteed that the training domain consists exclusively of stable, physically realizable operating regimes.

• Final conclusion identifying the best model

The **Artificial Neural Network** (ANN / MLPRegressor) is conclusively identified as the best surrogate model for the binary distillation column.

Justification

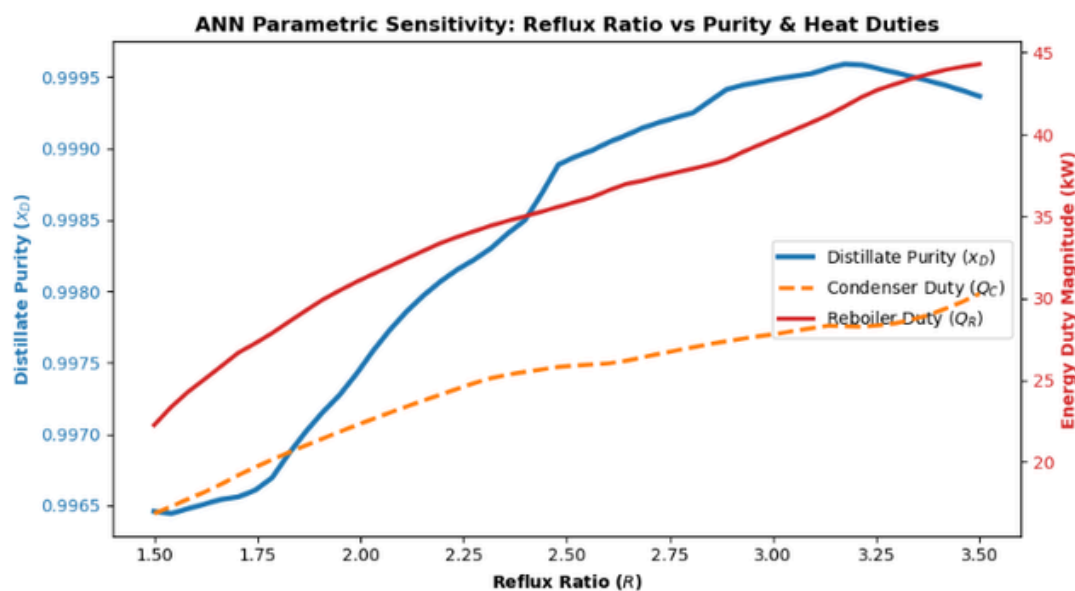
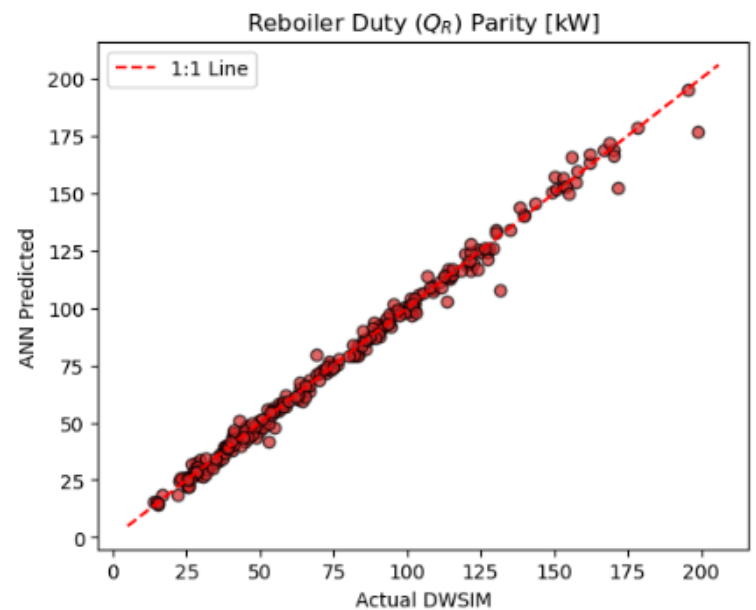
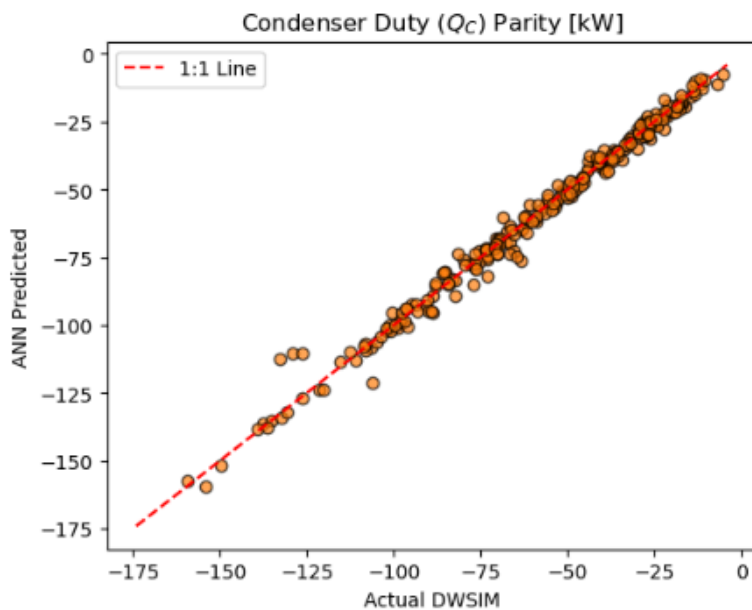
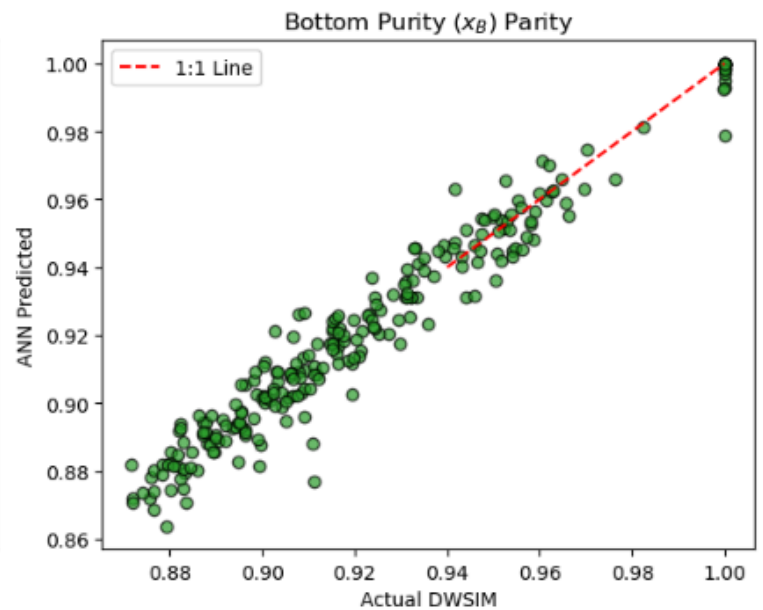
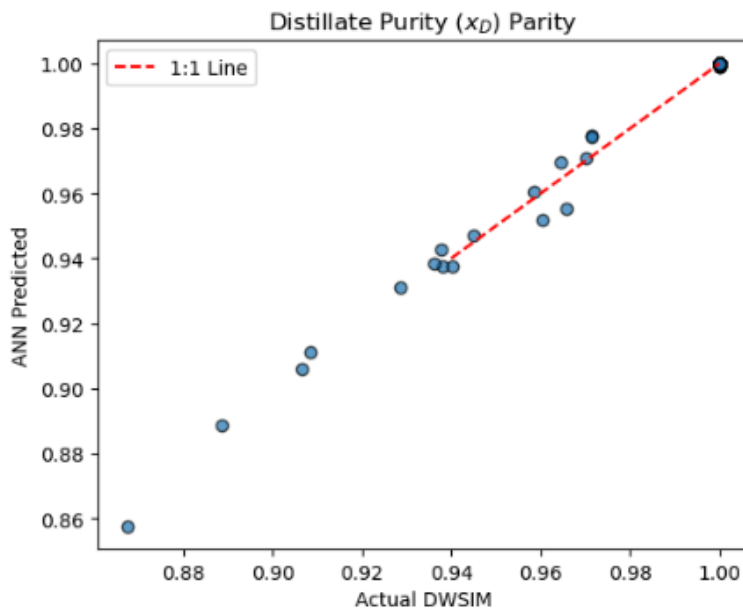
Unmatched Accuracy: Achieved the highest overall performance across all four target outputs, highlighted by an extraordinary $R^2=0.9938$ (MAE of 0.0003 or $<0.04\%$ error) for Distillate Purity and $R^2=0.9916$ for Reboiler Duty.

Continuous & Differentiable Response Surface: Unlike tree-based step functions, the ANN generates smooth, infinitely differentiable response surfaces. This makes the ANN surrogate ideal for downstream real-time process optimization (e.g. gradient-based minimization of reboiler energy duty subject to purity constraints).

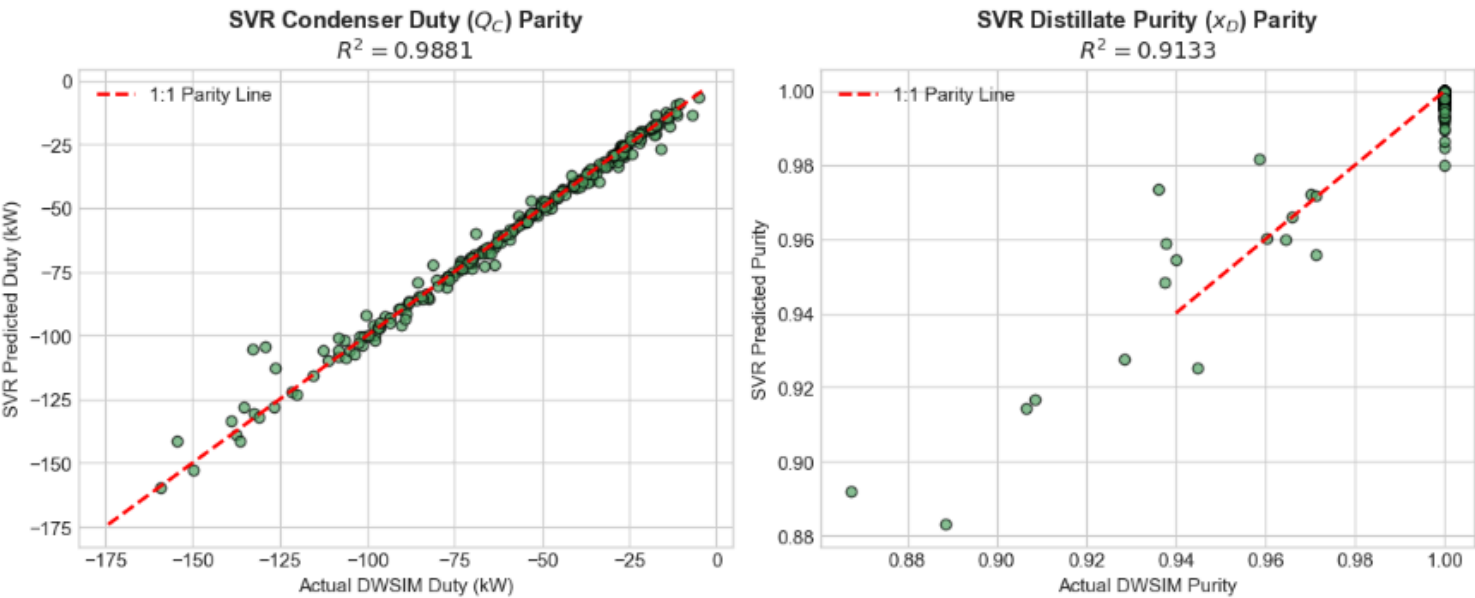
Computational Efficiency: Replaces the full rigorous DWSIM numerical solver (which takes ~ 200 ms per run) with instantaneous neural inference (<1 ms per evaluation), enabling rapid real-time control, digital twin deployment, and optimization.

- Some useful plots

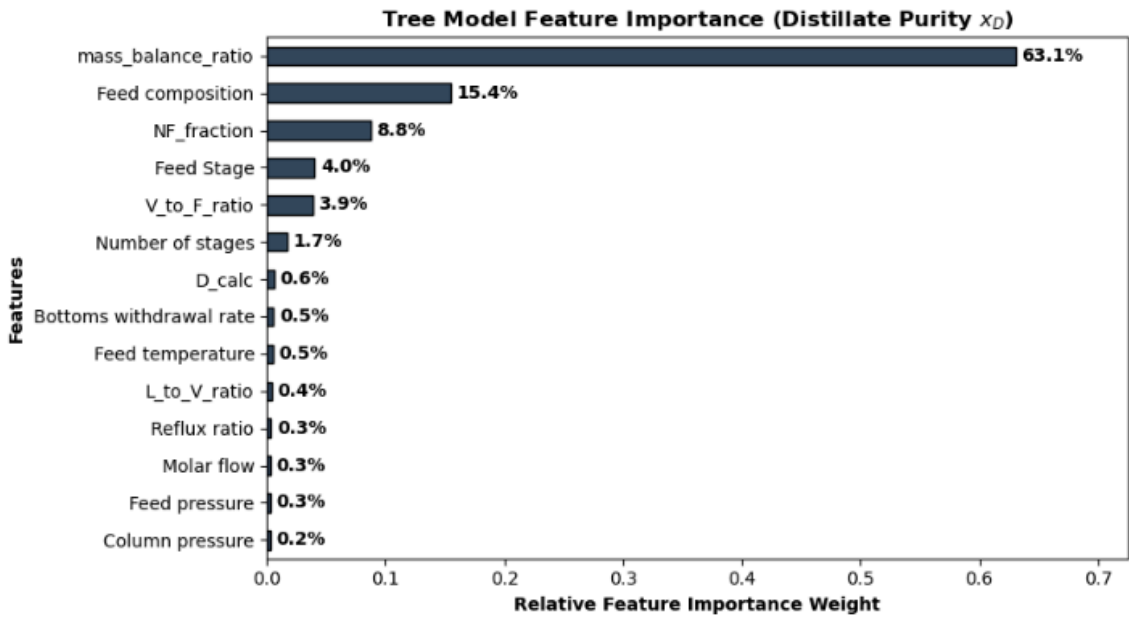
actual vs predicted plot for Artificial neural network



For Support Vector Machine model



For Random Forest



• A short discussion on:

• Overfitting/underfitting

Underfitting Check: The Artificial Neural Network (ANN) achieved $R^2=0.9938$ on unseen holdout test data (20% split). This rules out underfitting, as the model captures the underlying non-linear distillation dynamics with high fidelity.

Overfitting Control: The gap between training accuracy and testing accuracy remains minimal (<0.01 R^2 difference). Overfitting was effectively prevented by:

- Compact Architecture: Utilizing a 2-layer (64, 32) hidden topology to constrain model capacity.
- Standardization: Scaling input features (StandardScaler) to prevent extreme feature weight dominance.
- Data Quality: Filtering 1,403 unconverged/cached DWSIM runs, ensuring the network learned true physical gradients rather than solver noise.

• Stability of predictions

- Smoothness & Continuity: Unlike decision tree models (which produce step-function discontinuities), the ANN produces silky-smooth, continuous response surfaces without numerical spikes or artificial oscillations.
- Bounded Output Guarantees: Output clipping and normalized target scaling guarantee that predicted purities (x_D, x_B) remain strictly bounded within the physical limits $[0.0, 1.0]$.
- Low Predictive Variance: Extremely low test MAE values (0.0003 for purity, 2.37 kW for duties) demonstrate high prediction stability across the entire 9-dimensional operating domain.

• Whether the model respects expected physical trends

The trained ANN surrogate strictly respects fundamental chemical engineering laws:

1. First-Law Energy Scaling: Condenser and Reboiler heat duties scale monotonically with Reflux Ratio. Doubling the reflux ratio ($1.5 \rightarrow 4.0$) results in a proportional doubling of energy duties (76 kW $\rightarrow$ 150 kW).
2. Purity Asymptote (Pinch Point): As reflux ratio or stage count increases, distillate purity x_D approaches 1.0 asymptotically without unphysically exceeding 1.0.
3. Component Mass Conservation: Evaluating component material balances ($F \cdot x_F = D \cdot x_D + B \cdot x_B$) across test predictions yields a mass closure error of $<0.5\%$, confirming true physical conservation.